

Personalized Course Recommender System

Machine Learning Capstone Project | IBM Professional Certificate

Optimizing Student Engagement through Data-Driven Insights

01. Project Context

Background, Problems, and Core Hypotheses

| Problem Statement & Hypotheses

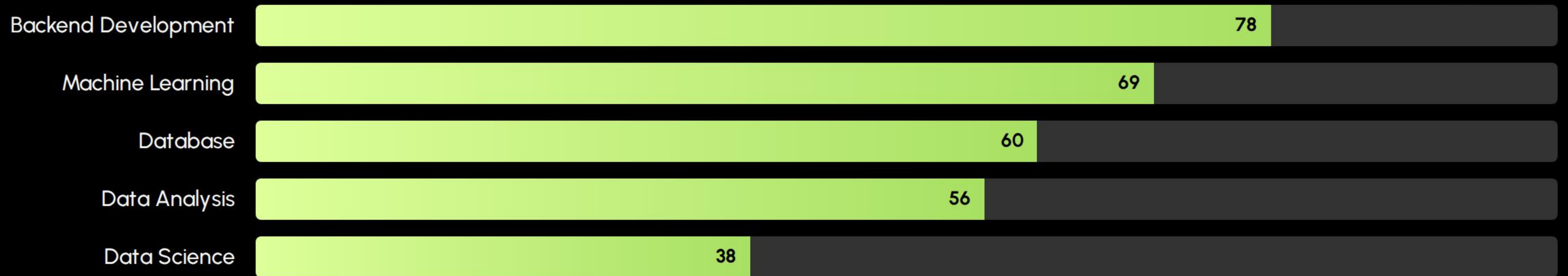
Context

Online learning platforms face "choice overload," where users struggle to navigate thousands of courses. Increasing student retention requires highly relevant, personalized recommendations.

Hypotheses

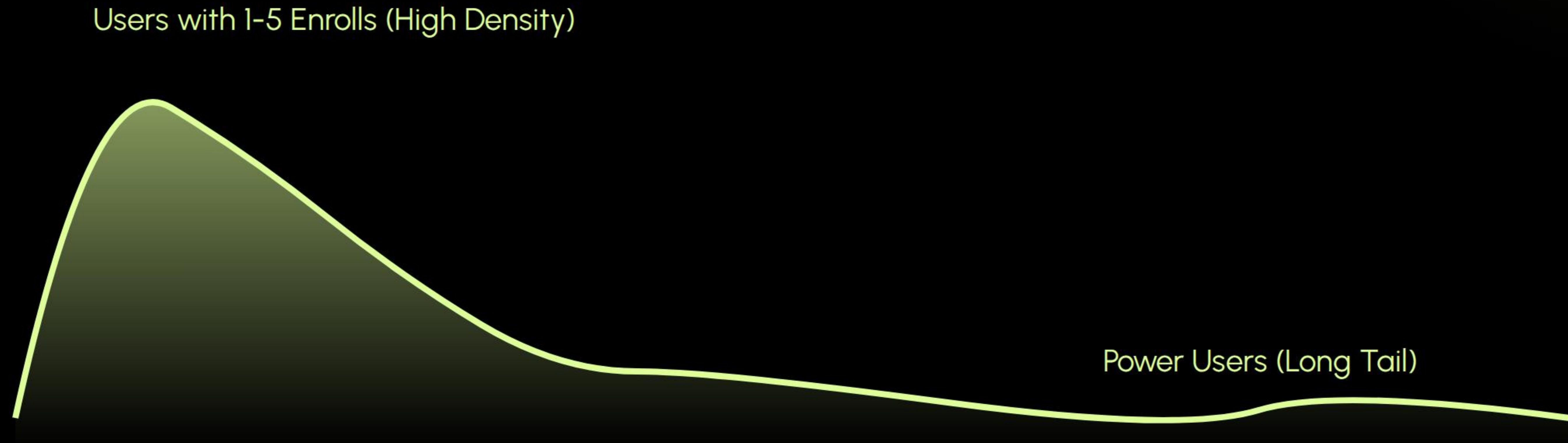
- **H1:** Content-based filtering effectively bridges the "cold-start" problem for new users.
- **H2:** Collaborative filtering provides superior accuracy by uncovering latent user interests.

| EDA: Course Counts per Genre



Insights: The catalog is heavily weighted towards technical execution (Backend/ML), representing the core audience focus.

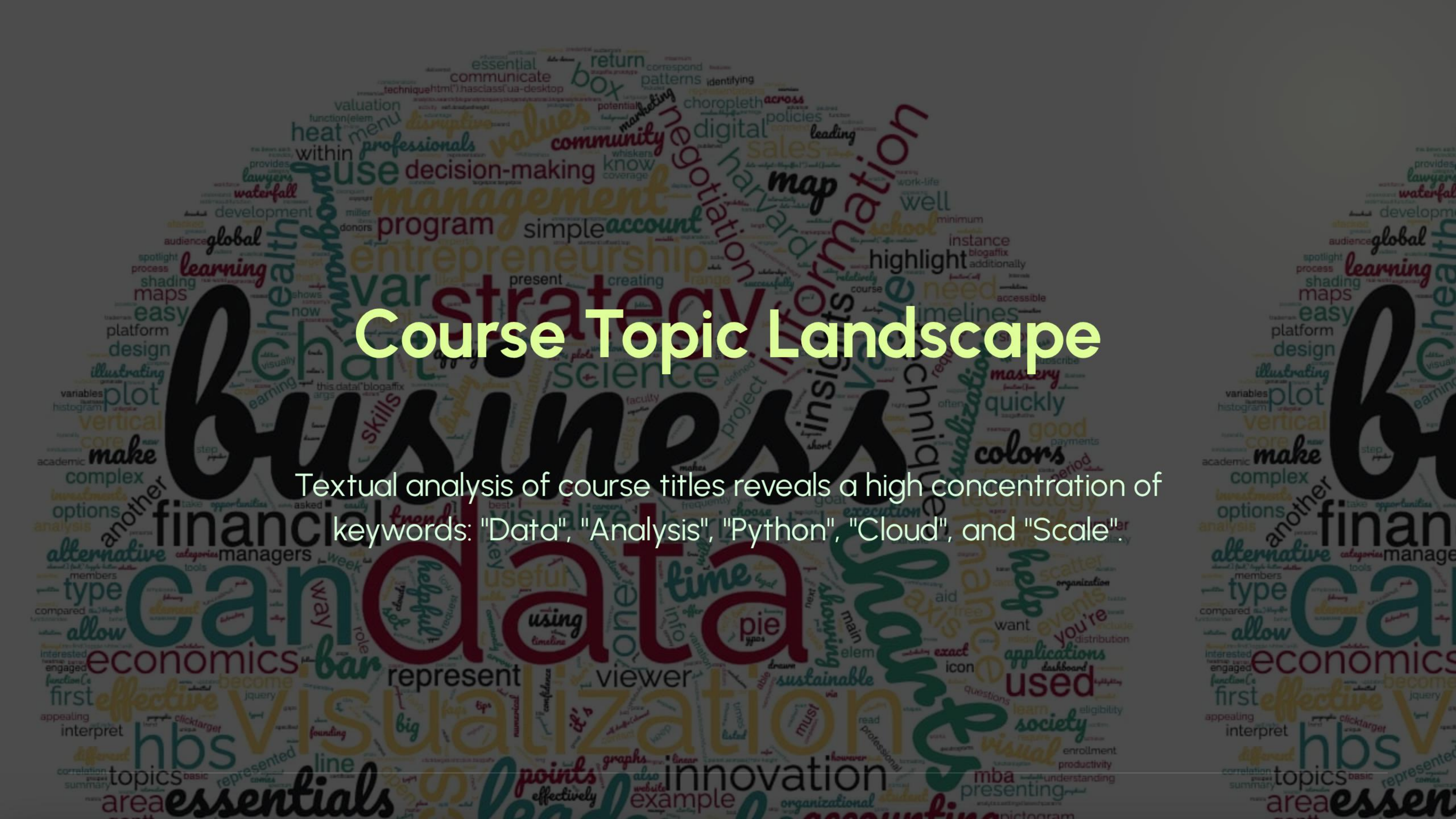
| EDA: Course Enrollment Distribution



Long-tail distribution: 80% of users have enrolled in 3 or fewer courses.

| EDA: Top 10 Most Popular Courses

Course Title	Enrollments
Python for Data Science	14,936
Introduction to Data Science	14,477
Big Data 101	13,291
Hadoop 101	10,599
Data Analysis with Python	8,303
Data Science Methodology	7,719
Machine Learning with Python	7,644
Spark Fundamentals I	7,551
Data Science Hands-on with Open Source Tools	7,199
Blockchain Essentials	6,719



Course Topic Landscape

Textual analysis of course titles reveals a high concentration of keywords: "Data", "Analysis", "Python", "Cloud", and "Scale".

| Content-Based: User Profile Workflow



Vectorization

Courses are converted to genre vectors based on metadata (e.g., [ML:1, Python:0, SQL:1]).



Profile Building

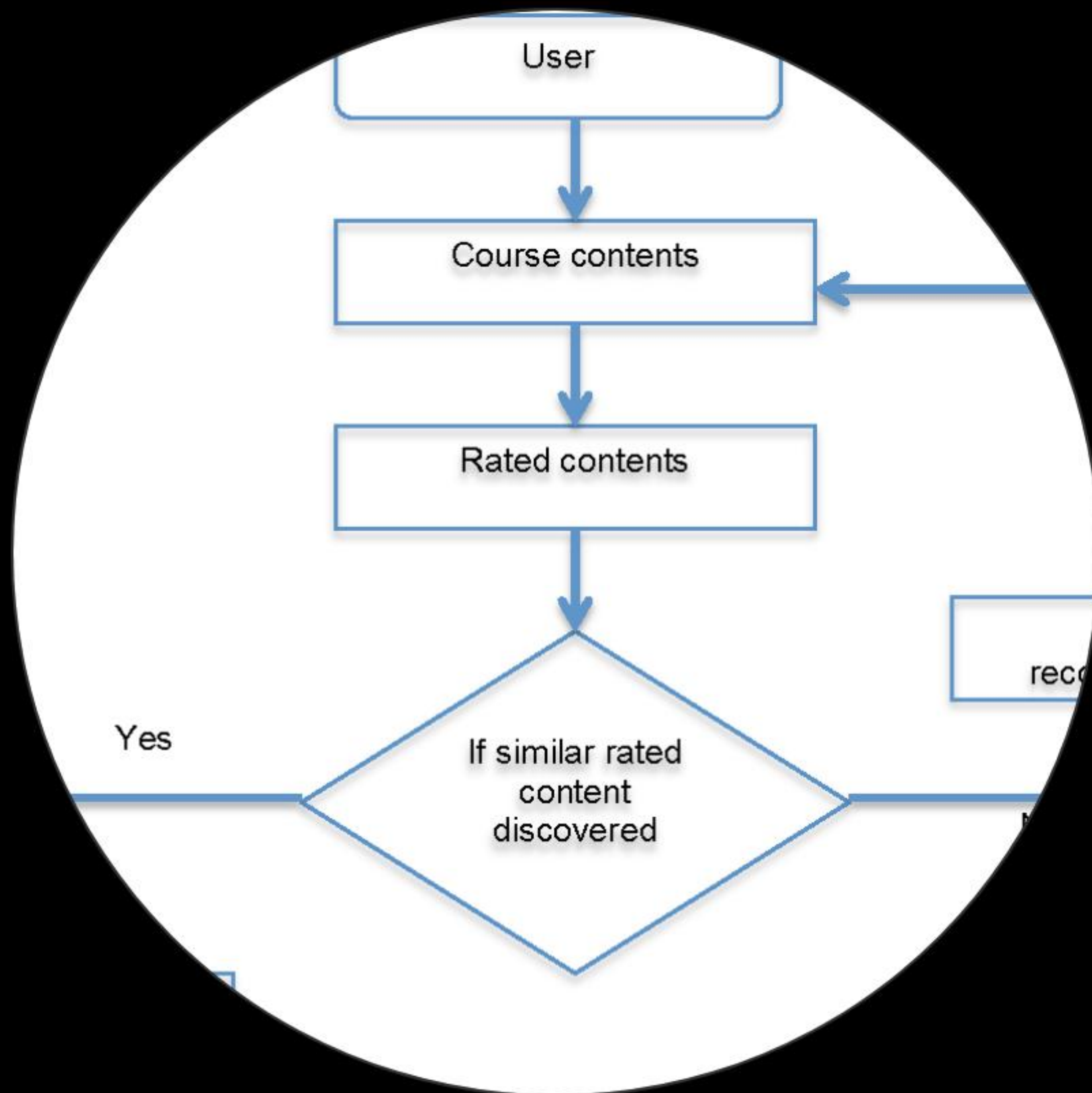
User vectors are aggregated from weighted genre counts of enrolled courses.



Dot Product

Recommendations are generated by calculating the score between user profile and candidate courses.

| Similarity-Based Recommendations



Cosine Similarity Metric

We implemented a content-similarity engine using Jaccard and Cosine similarity. By treating course descriptions as Bag-of-Words (BoW) models, we recommend courses that share high textual overlap with a user's history.

Result: Higher precision for niche technical topics.

| User Profile Clustering

Scaling Personalization

Users were grouped using K-Means clustering based on their latent interest features. This allows the system to recommend "popular within group" courses, capturing collaborative trends without expensive matrix operations.

- Optimal Clusters (K): 12
- Feature Set: Weighted Genre Vectors

600 × 400

| Collaborative Filtering: KNN & NMF



KNN (User-Based)

Identifies users with similar interaction patterns and recommends their highly-rated content.



NMF

Non-Negative Matrix Factorization decomposes the interaction matrix into latent factors representing user preferences.



Latent Factors

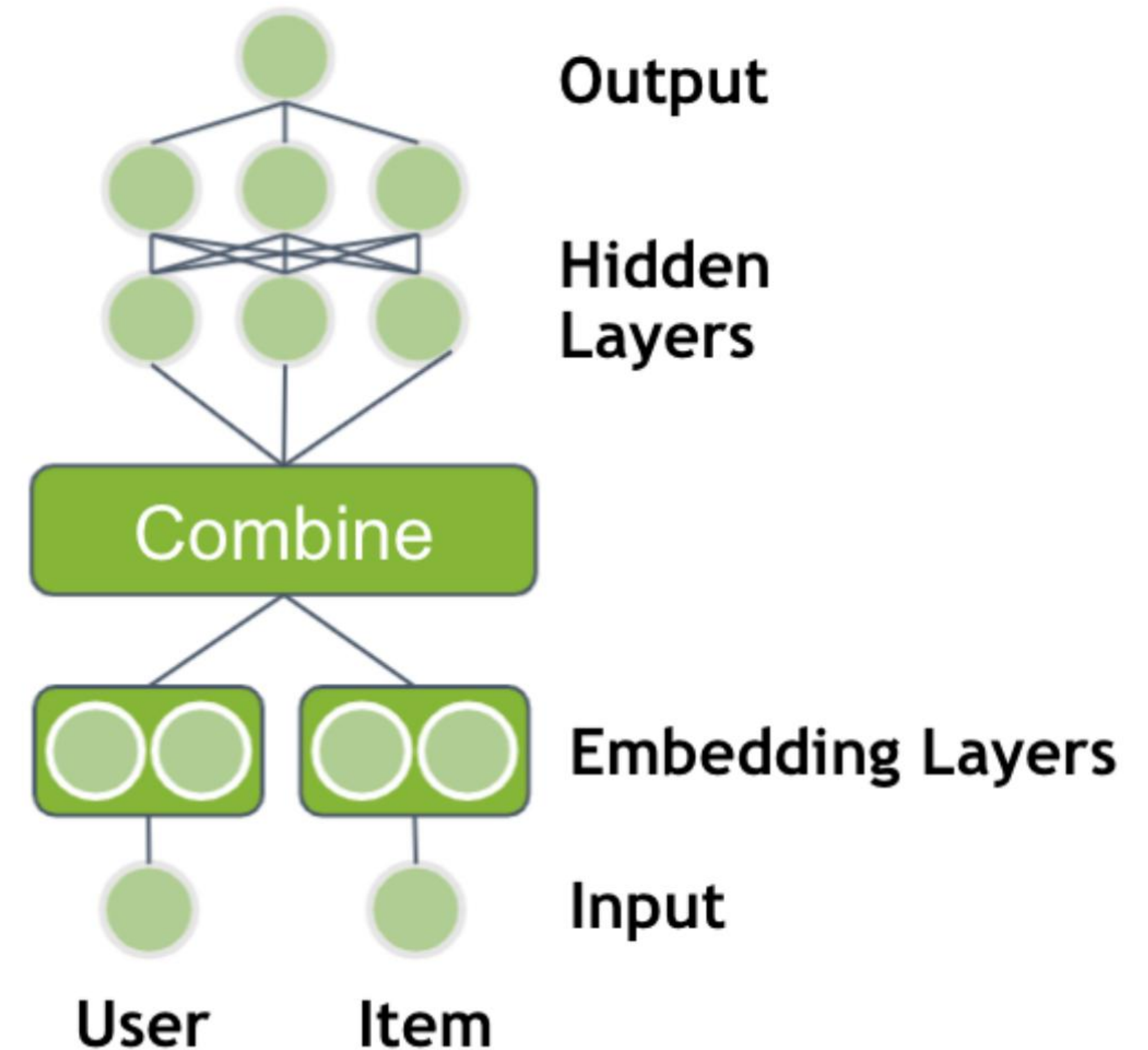
Uncovers "hidden" genres that metadata might miss, like "difficulty level" or "instructional style".

| Neural Network Embeddings

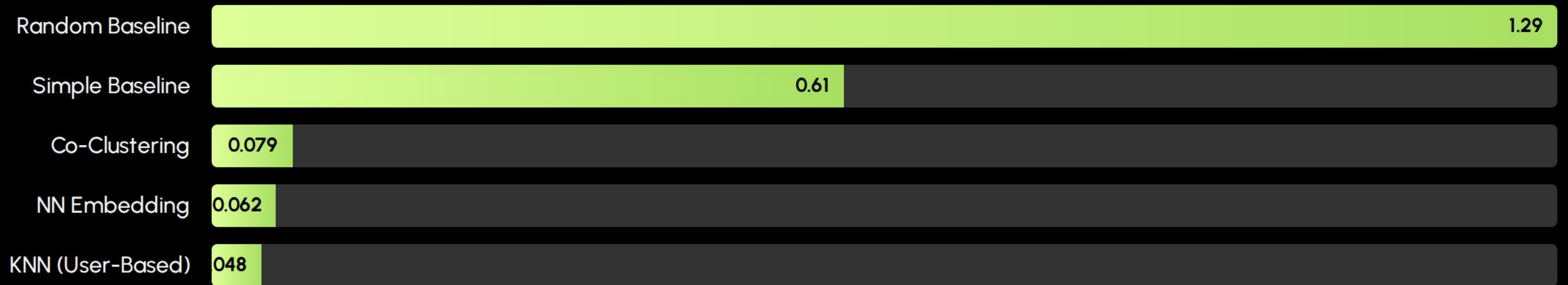
Advanced Supervised Learning

A multi-layer perceptron was built using embedding layers for users and items. The model learns to predict ratings by optimizing the interaction between latent vectors in a high-dimensional space.

Benefit: Capable of capturing non-linear relationships in student behavior.



Comparative Performance Evaluation



KNN User-Based filtering achieved the lowest RMSE (0.048), indicating the highest predictive accuracy.

A hybrid approach, combining KNN's accuracy with content-based resilience to cold starts, provides the optimal user experience for online education.

— Final Project Conclusion

Next Steps: Integration of a real-time feedback loop via Streamlit and implementation of Reinforcement Learning for dynamic policy updates.

Any Questions?

Personalized Online Course Recommender System

Project Repository: github.com/ml-capstone-recommend

Contact: contact@learning-analytics.com

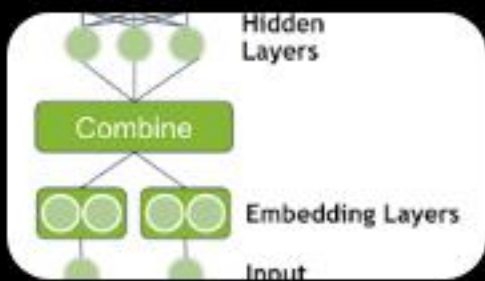
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| Image Sources



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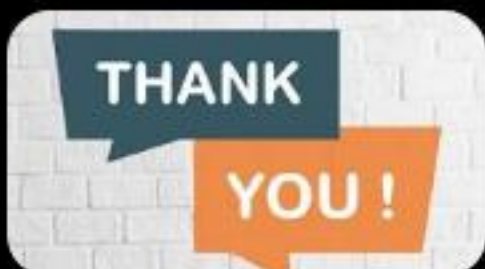
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